COMBINATIONAL CIRCUITS

Combinational logic

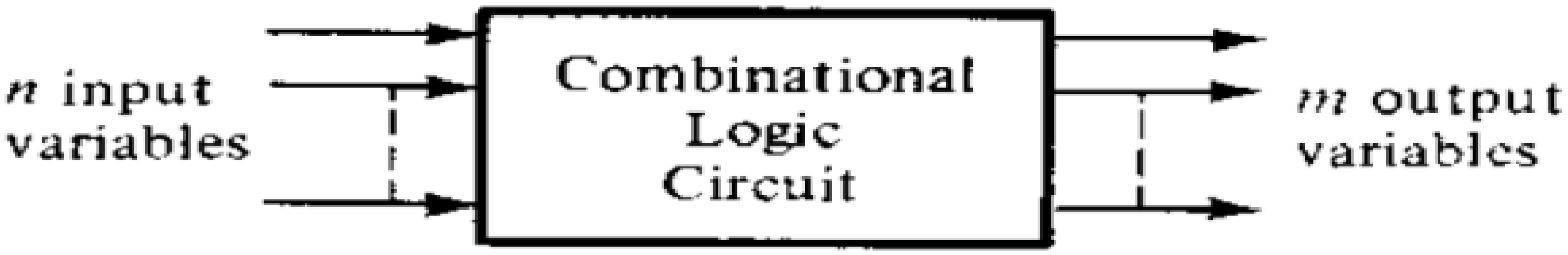
- A combinational circuit consists of logic gates whose outputs at any time are determined directly from the present combination of inputs without regard to previous outputs.

- A combinational circuit performs a specific information processing operation fully specified logically by a set of Boolean functions.

- A Combinational circuit consists of input variables, logic gates, and output variables.

Combinational logic

- The logics gate accept signals from the inputs and generate signals to the outputs.
- This process transforms binary information from the given input data to the required output data.



Combinational logic

- Design Procedures
  - Starts from the verbal outline of the problem and ends in a logic circuit diagram.
  - The procedure involves the following step,
    - The problem is stated.
    - Input and required output variables are determined.
    - Assigned the variables letter symbols.
    - Make the truth table.
    - The simplified Boolean functions for each output is obtained.
    - The logic diagram is drawn.

Combinational logic

- Adders
  - Adders are important in computers and also in other types of digital systems in which numerical data are processed.
  - An understanding of the basic adder operation is fundamental to the study of digital systems.
  - The most basic operation is no doubt is the addition of two binary digits.

The Half Adder

- Half Adder
  - The combinational circuit that performs the additions of two bit is called Half adder.
  - One that performs the addition of three bits including two digits and one previous carry is a full adder.
  - Two half adders can be employed to form a full adder.

Combinational logic

- **Half Adder**
  - It has two inputs and two outputs.
  - The input variables designates the augends and addend bits; the output variables produces the sum and carry.
  - It is necessary to specify two output variables because the result may consist of two binary digits.
  - A and B are two inputs binary variables while C and S used for carry and Sum to the outputs.

Combinational logic

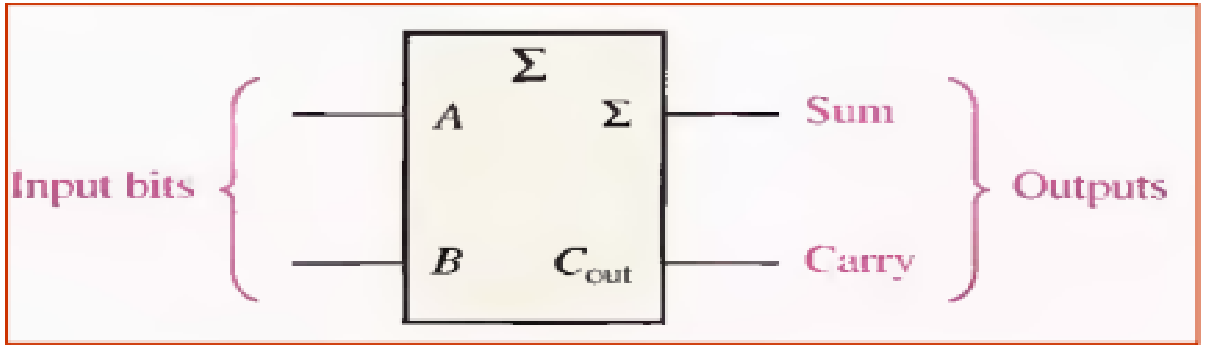
- **Half Adder**
  - The half-adder accepts two binary digits on its inputs and produces two binary digits on its outputs, a sum bit and a carry bit.
  - The truth table look like this,

$$
\begin{aligned}
0 + 0 &= 0 \\
0 + 1 &= 1 \\
1 + 0 &= 1 \\
1 + 1 &= 10
\end{aligned}
$$

A	B	C_{out}	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Combinational logic

- Half Adder
  - A half-adder is represented by the logic symbol in Figure below,



Half Adder

- Half-Adder Logic
  - Notice that the output Carry (C_{out}) is a 1 only when both A and B are 1s; therefore C_{out} can be expressed as the AND of the input variables.
  - $C_{out} = AB$
  - Now observe that the sum output (Σ) is a 1 only if the input variables. A and B, are not equal.
  - The sum can therefore be expressed as the exclusive-OR of the input variables.

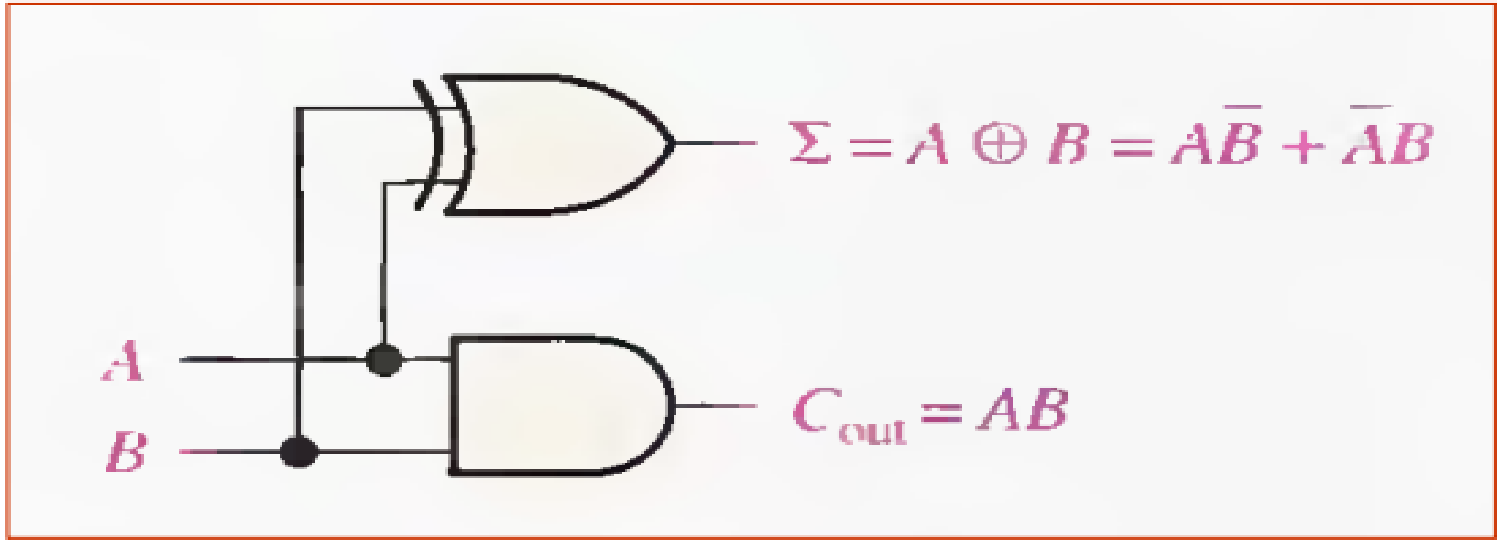
$$\Sigma = A \oplus B$$

Related Example

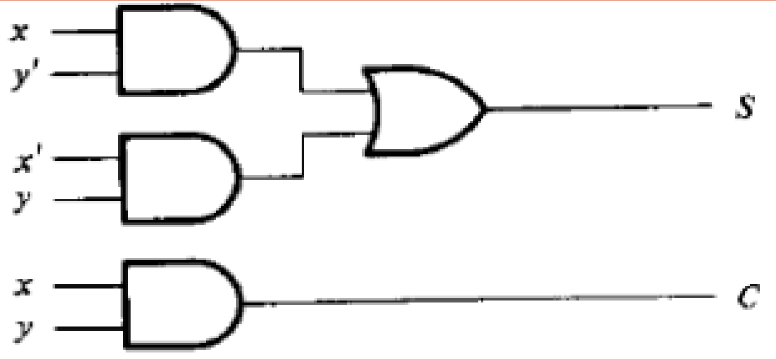
- Half-Adder Logic
  - The logic implementation required for the half adder function can be developed.
  - The output carry is produced with an AND gate with A and B on the inputs.
  - The sum output is generated with an exclusive-OR gate.

Half Adder

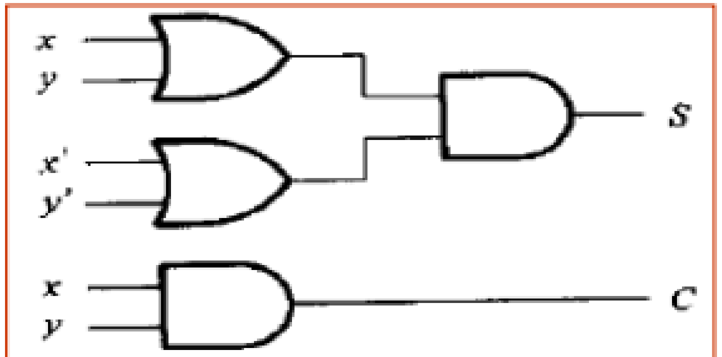
- Half-Adder Logic diagram



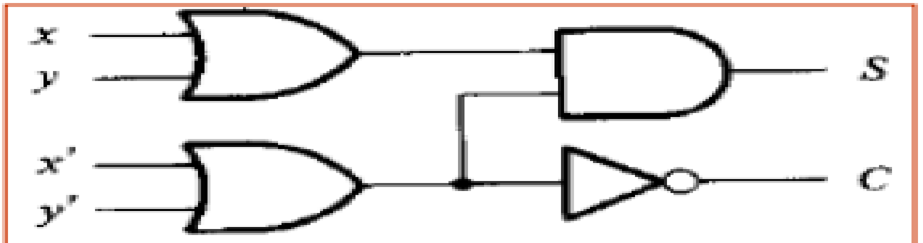
Solutions



(a) $S = xy' + x'y$
$C = xy$



(b) $S = (x + y)(x' + y')$
$C = xy$



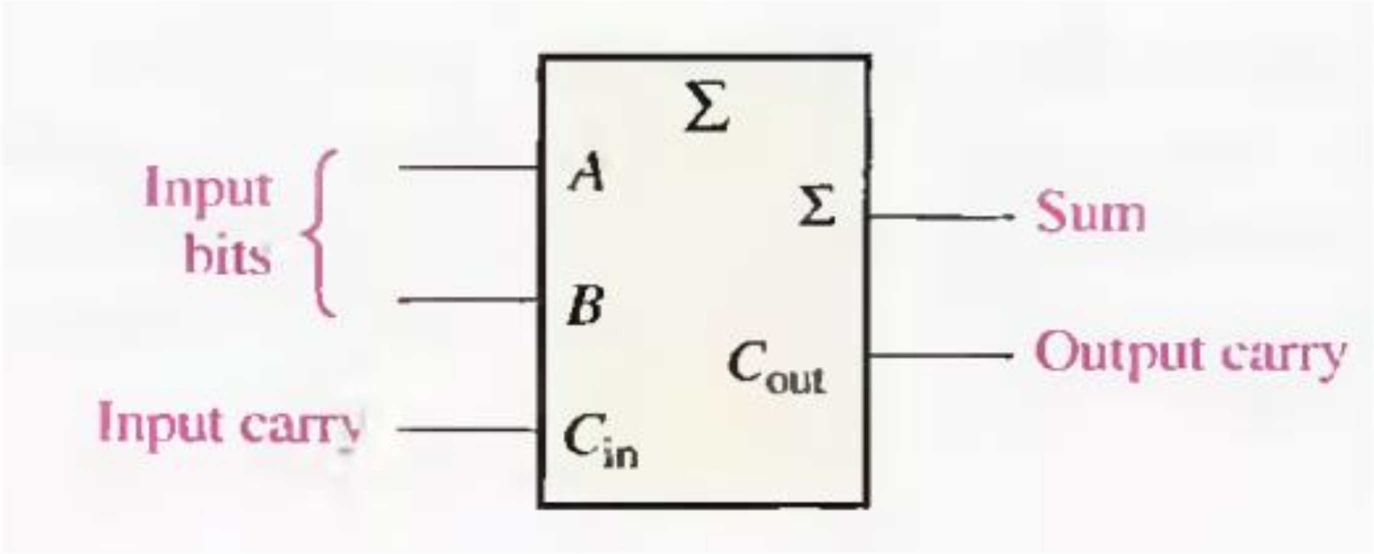
(d) $S = (x + y)(x' + y')$
$C = (x' + y')'$

Full Adder

- Full Adder
  - The second category of adder is the full-adder.
  - The full-adder accepts two input bits and an input carry and generates a sum output and an output carry.
  - The basic difference between a full-adder and a half-adder is that the full-adder accepts an input carry.

Full Adder

- Logical symbol for full adder is,



Full Adder

Truth Table

A	B	C_{in}	C_{out}	Σ
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

C_{in} = input carry, sometime designated as CI
C_{out} = output carry sometimes designated as CO
Σ=sum
A and B = input variables (operands)

Full Adder

- Full Adder Logic
  - The full-adder must add the two input bits and the input carry.
  - From the half-adder you know that the sum of the input bits A and B is the exclusive-OR of those two variables, A xor B.
  - For the input carry (C_{in}) to be added to the input bits. it must be exclusive-ORed with A xor B, yielding the equation for the sum output of the full-adder.

$$\Sigma = (A \oplus B) \oplus C_{in}$$

Full Adder

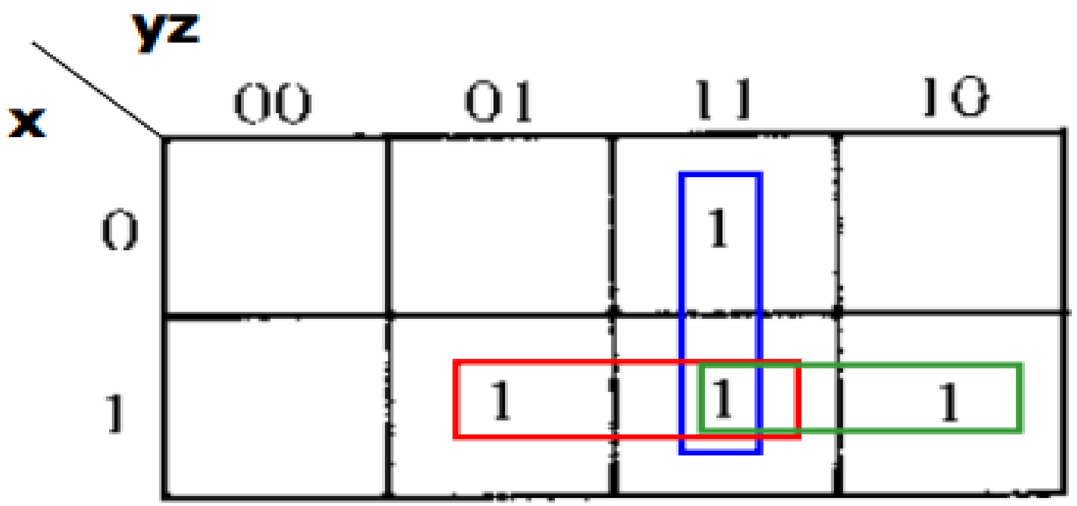
- Map for Full Adder (For Sum Function)

x \ yz	00	01	11	10
0		1		1
1	1		1	

S= x'y'z + x'yz' + xy'z' + xyz

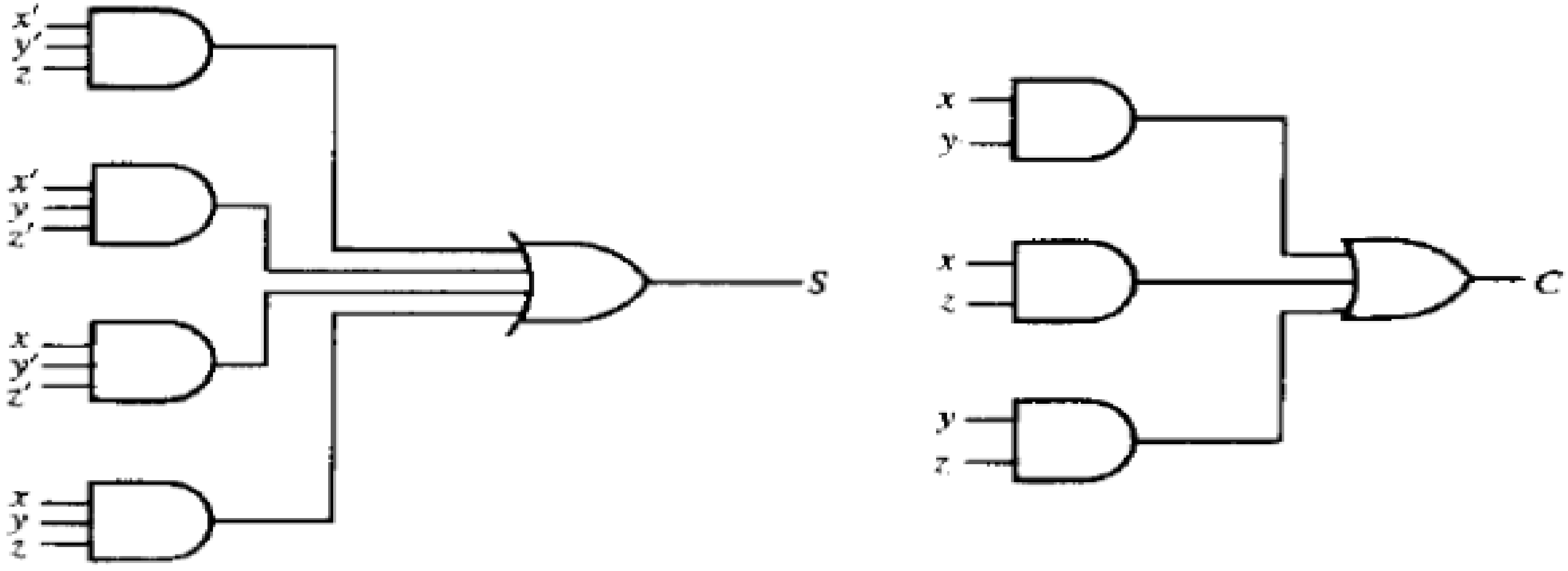
Full Adder

For Carry Simplified Expression



$$C = xy + xz + yz$$

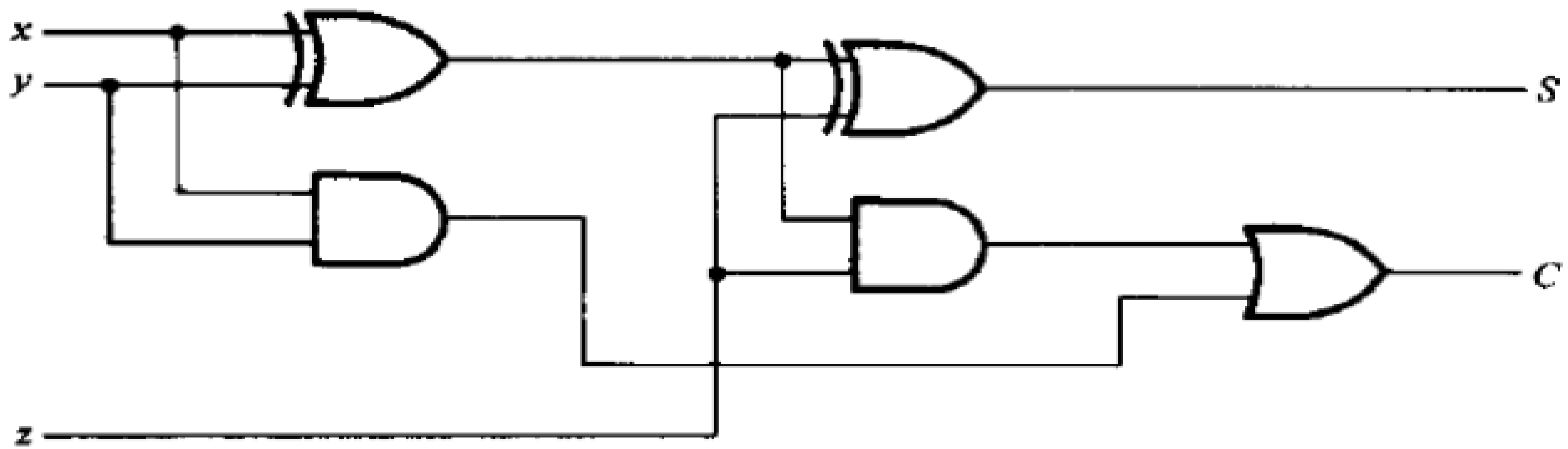
Implementation of Full Adder in SOP



Logic Diagram

Implementation of Full Adder

Implementation of a full adder with two half adders and an OR Gate



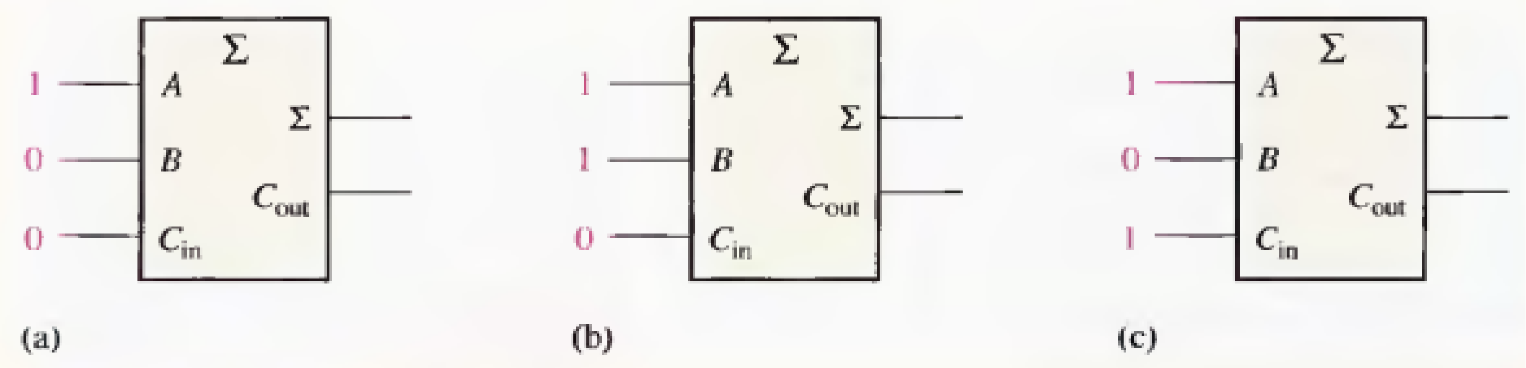
Adders

$$S = z \oplus (x \oplus y)$$
$$= z'(xy' + x'y) + z(xy' + x'y)'$$
$$= z'(xy' + x'y) + z(xy + x'y')$$
$$= xy'z' + x'yz' + xyz + x'y'z$$

$$C = z(xy' + x'y) + xy = xy'z + x'yz + xy$$

Problem

- For each of the three full-adders in Figure below, determine the outputs for the inputs shown.



Solution

(a) The input bits are $A = 1$, $B = 0$, and $C_{in} = 0$.

$$1 + 0 + 0 = 1 \text{ with no carry}$$

Therefore, $\Sigma = 1$ and $C_{out} = 0$.

(b) The input bits are $A = 1$, $B = 1$, and $C_{in} = 0$.

$$1 + 1 + 0 = 0 \text{ with a carry of } 1$$

Therefore, $\Sigma = 0$ and $C_{out} = 1$.

(c) The input bits are $A = 1$, $B = 0$, and $C_{in} = 1$.

$$1 + 0 + 1 = 0 \text{ with a carry of } 1$$

Therefore, $\Sigma = 0$ and $C_{out} = 1$.

THANK YOU